

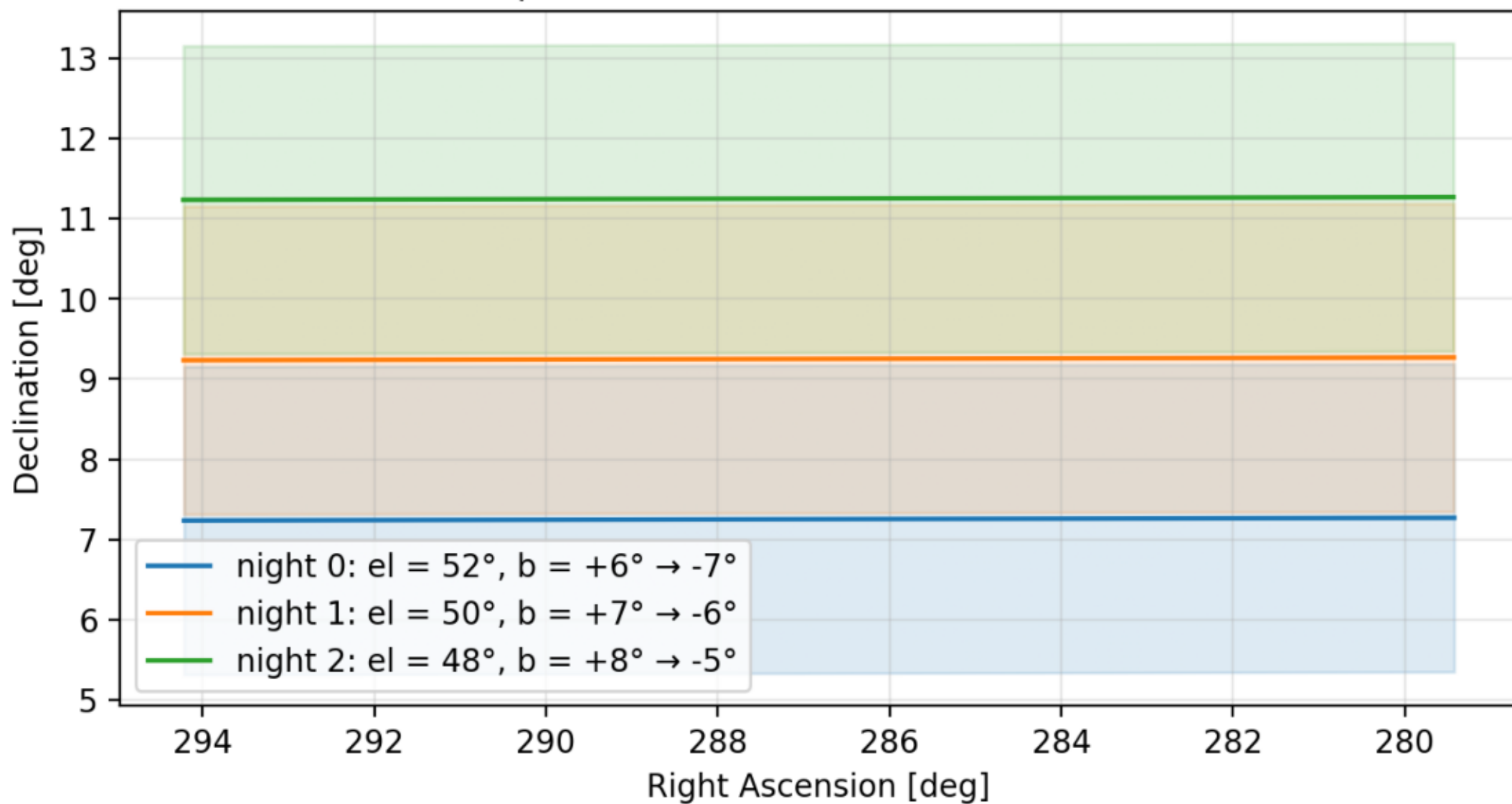
Experiment 003 — galplane-drift-sidelobe-stress

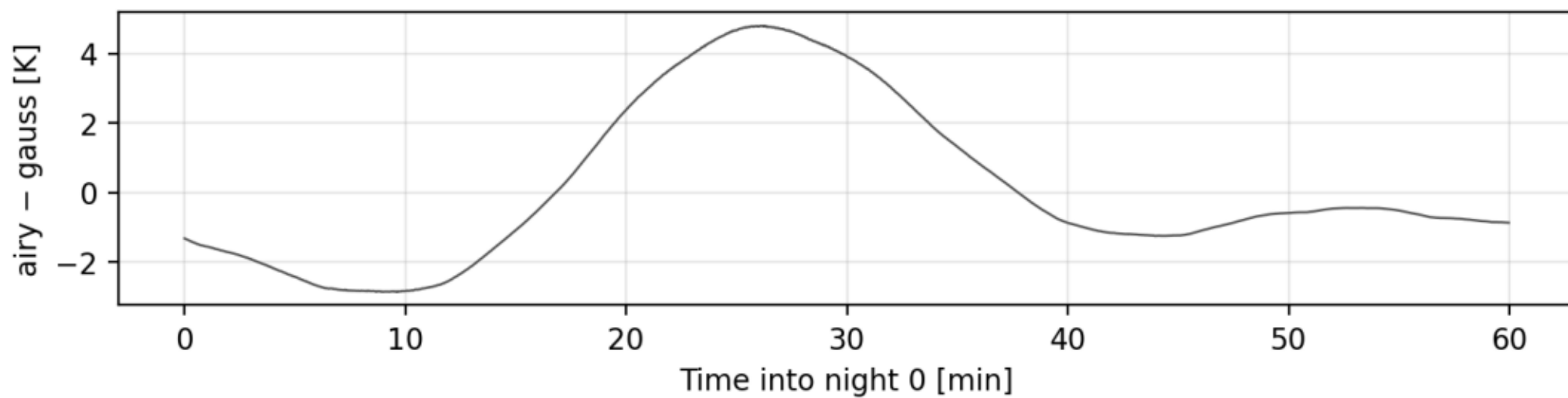
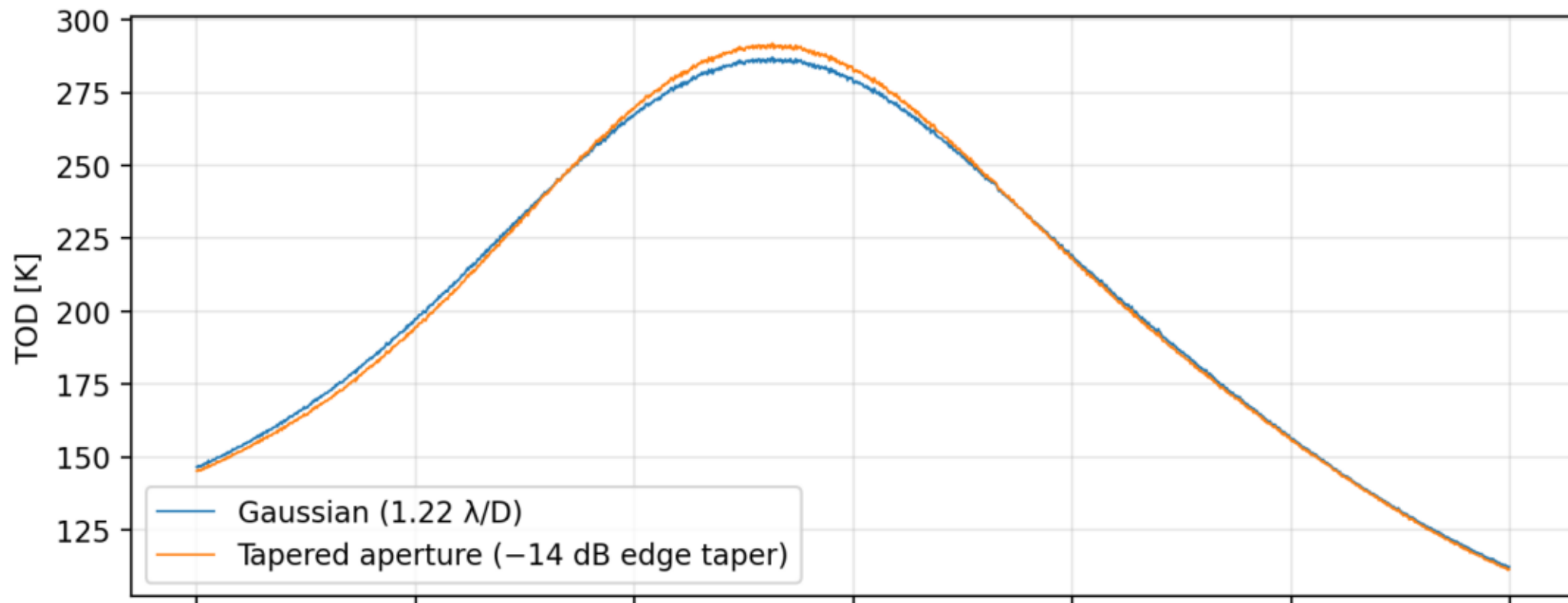
2026-07-23

Galactic-plane sidelobe stress test: identical drift design, beams, noise seeds and map-making recipe as experiments 001/002, but the LST window (start 2024-04-16 03:34:00 UTC, LST $\approx 280^\circ$) sweeps the Dec $\approx +9^\circ$ strip across the galactic plane ($b = +7^\circ \rightarrow -6^\circ$, crossing at $l \approx 43^\circ$). Scenarios: matched Gaussian, matched tapered aperture, and beam mismatch (tapered-aperture sky through the Gaussian operator). The mismatch — gauss rows isolate the unmodelled-sidelobe map error (same operator, mask, prior and noise realization).

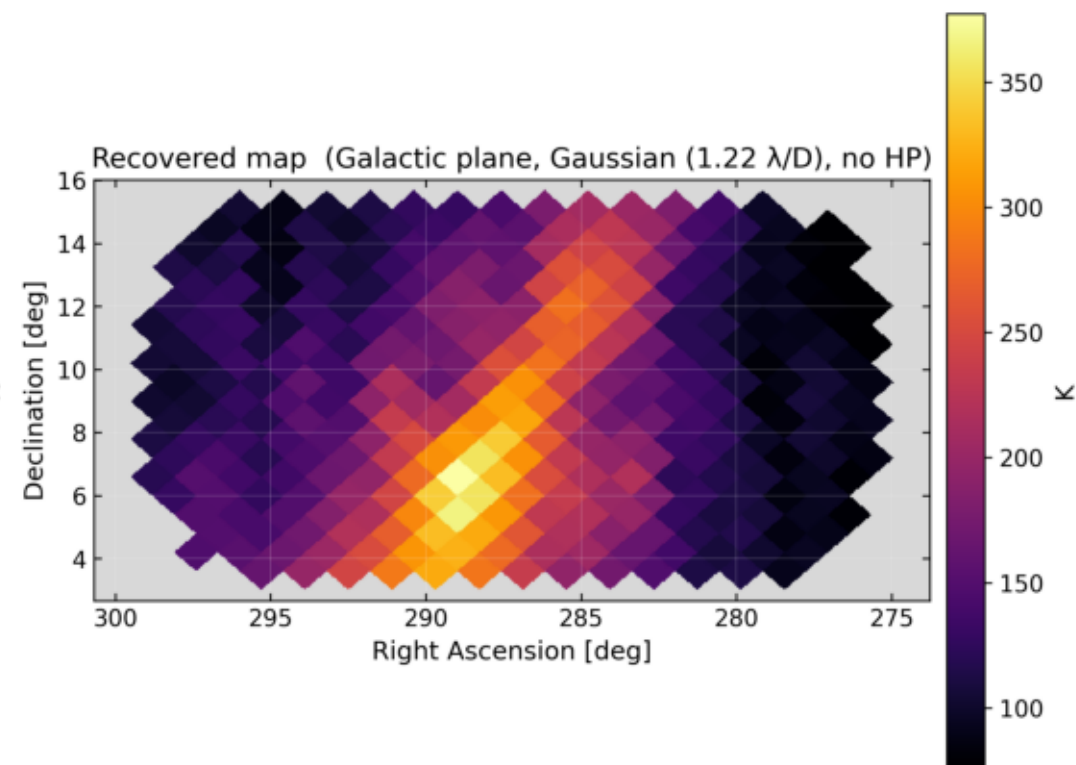
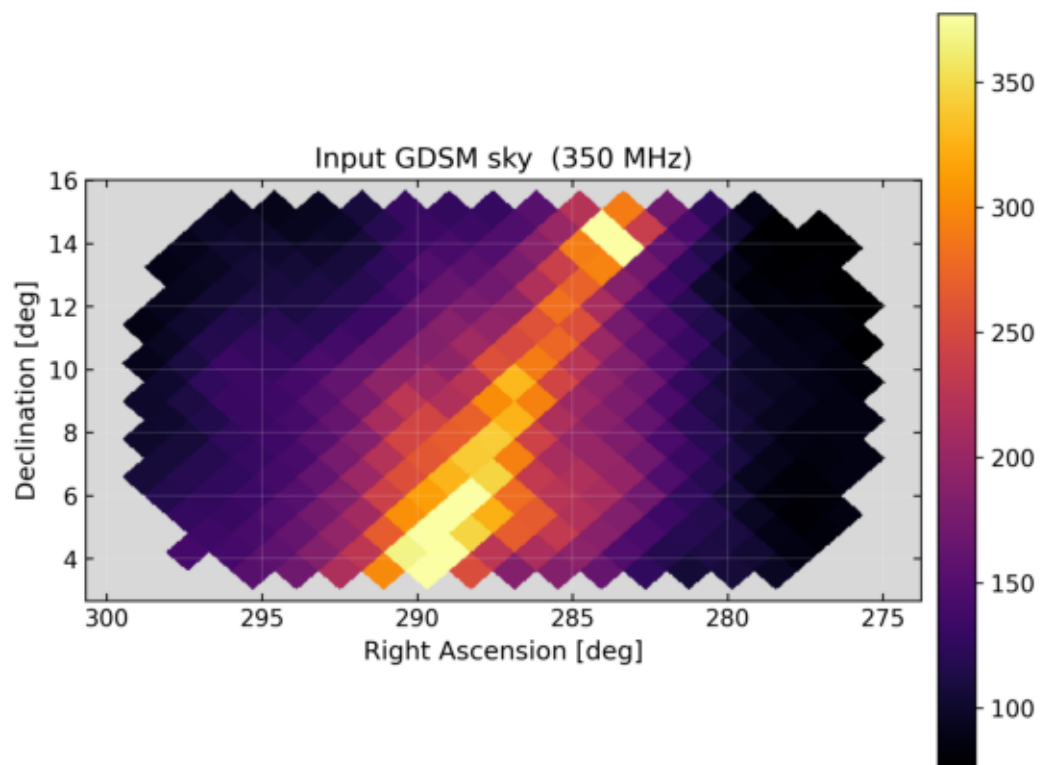
gauss / no HP	23081.4 mK
gauss / HP 2 mHz	31439.3 mK
airy / no HP	25262.0 mK
airy / HP 2 mHz	32206.9 mK
mismatch / no HP	23901.2 mK
mismatch / HP 2 mHz	30668.1 mK
mismatch — gauss / no HP	7975.3 mK
mismatch — gauss / HP 2 mHz	4496.9 mK

Galactic-plane drift tracks (bands = beam FWHM)

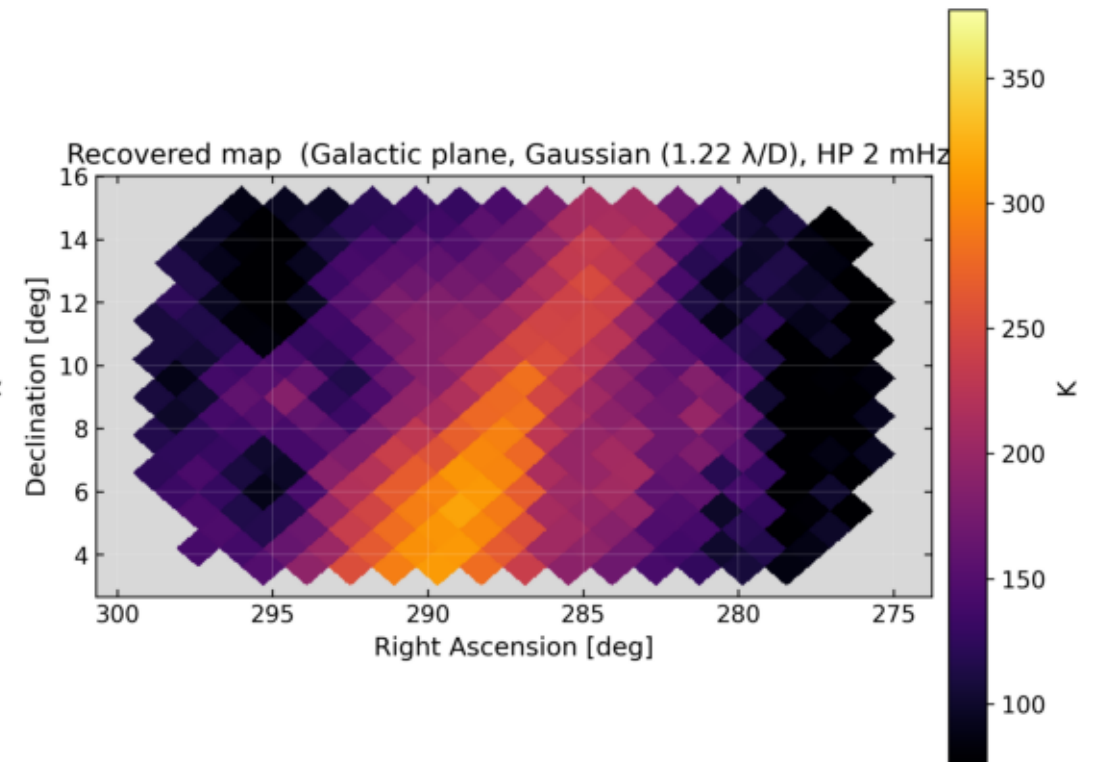
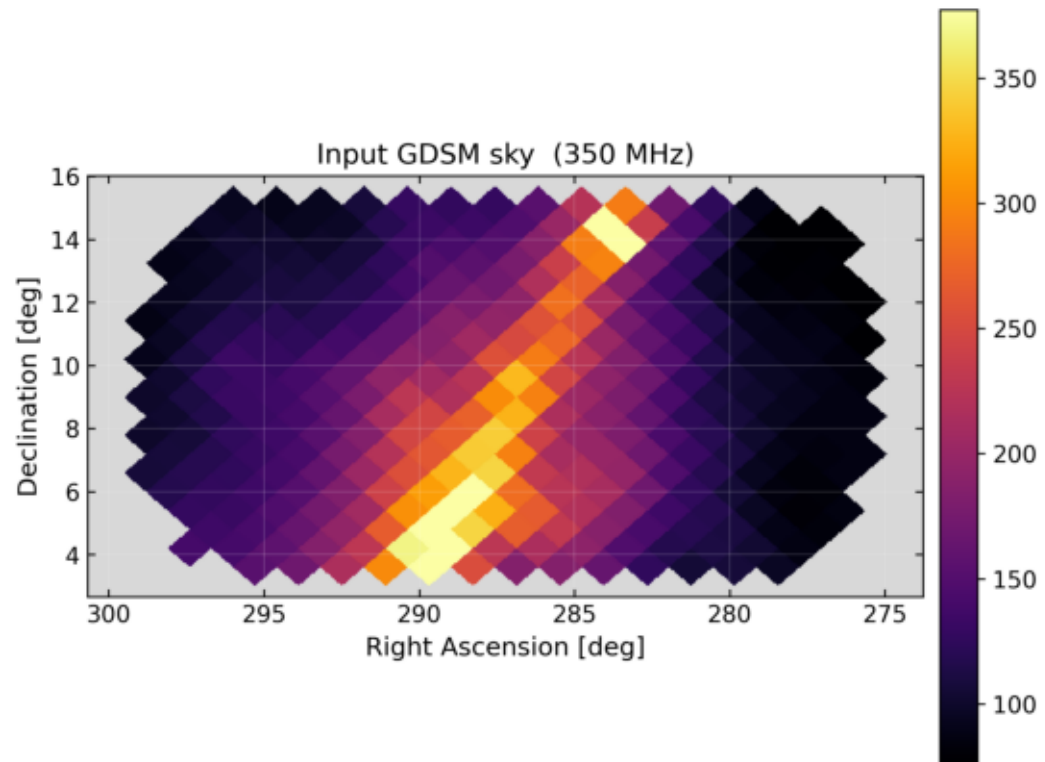




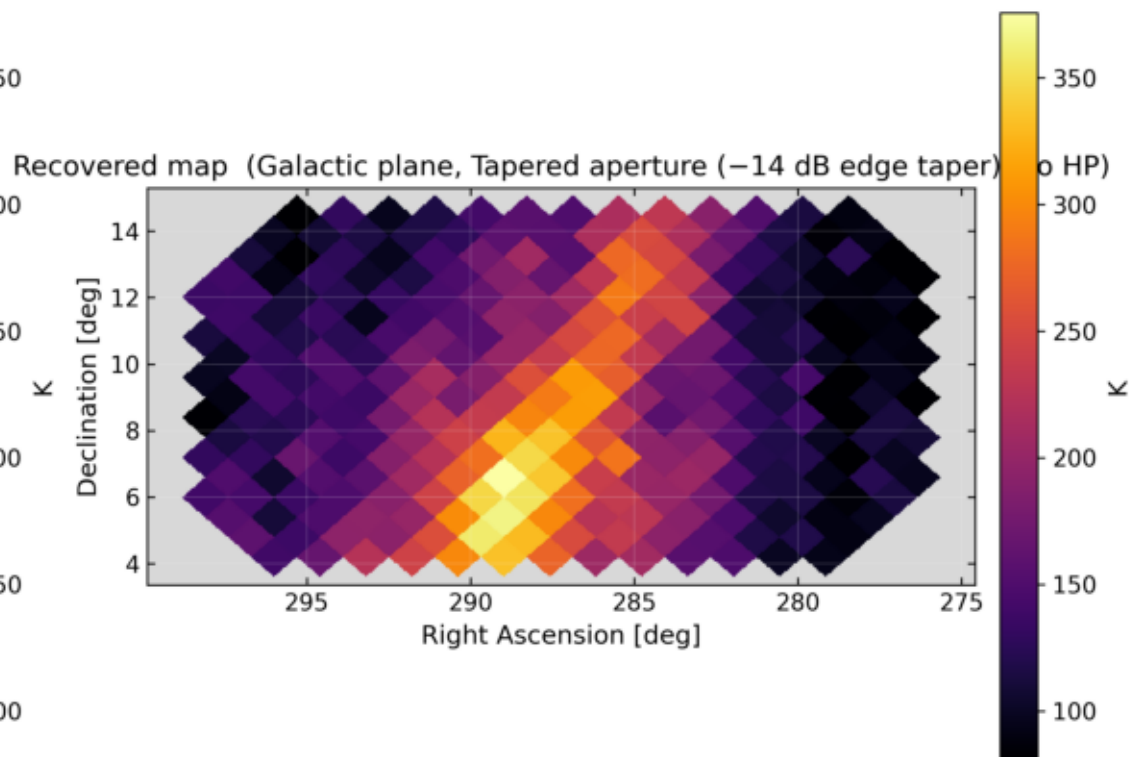
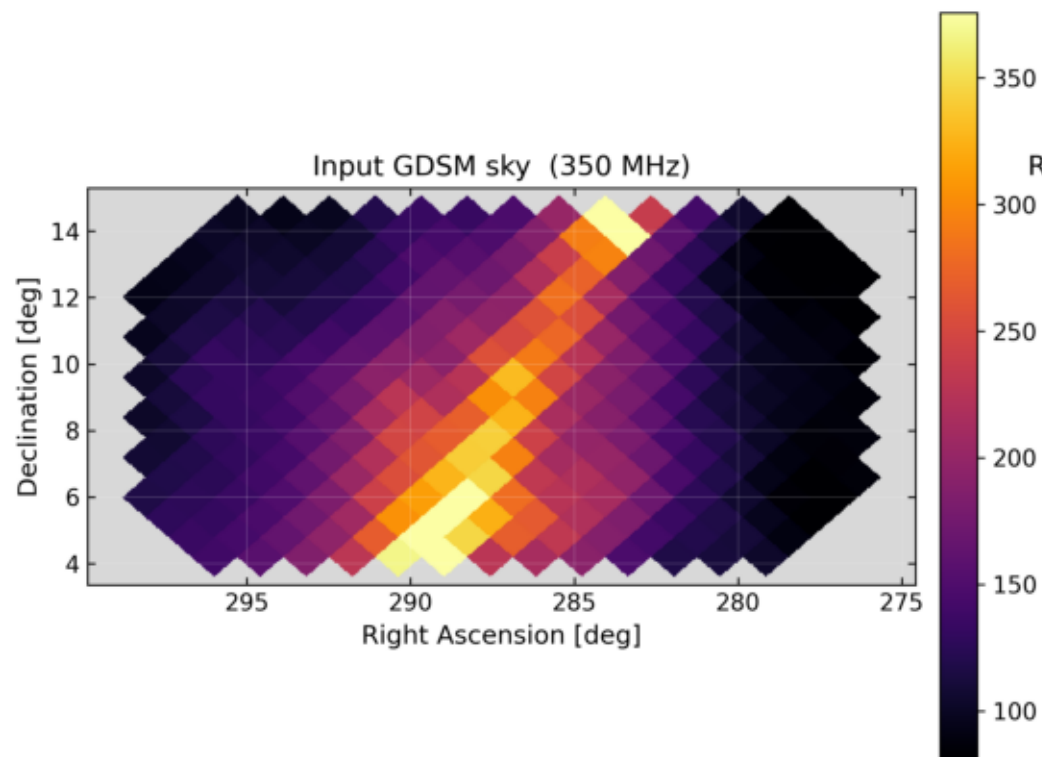
map_ska_galplane_gauss_noHP.png



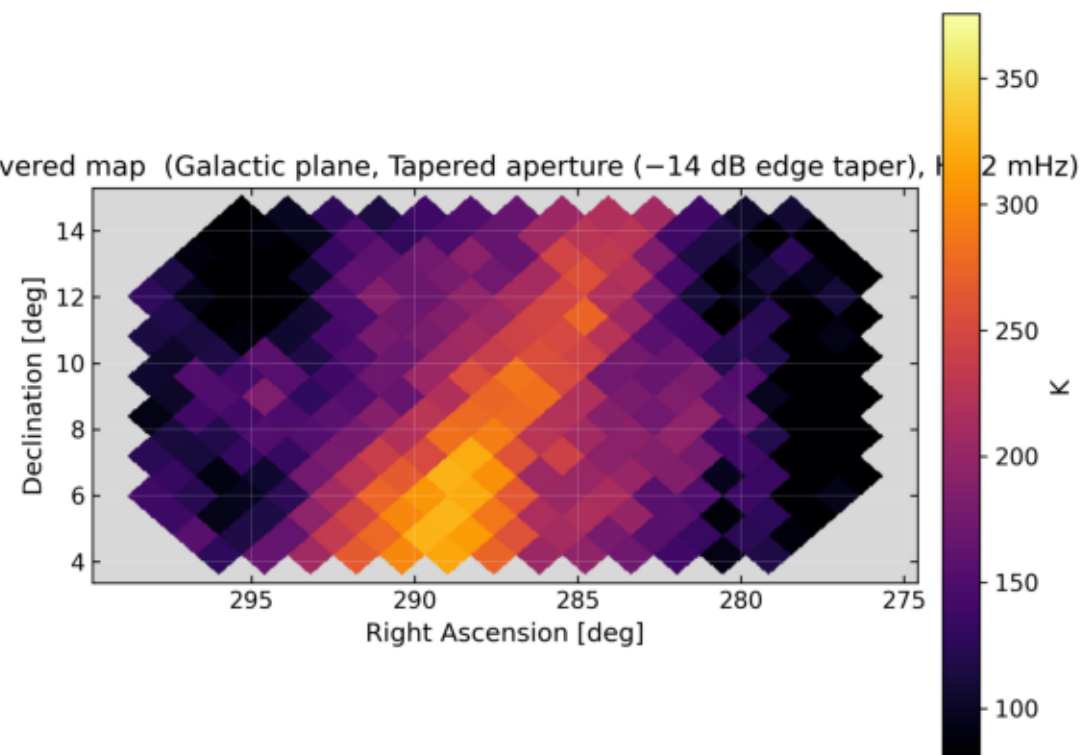
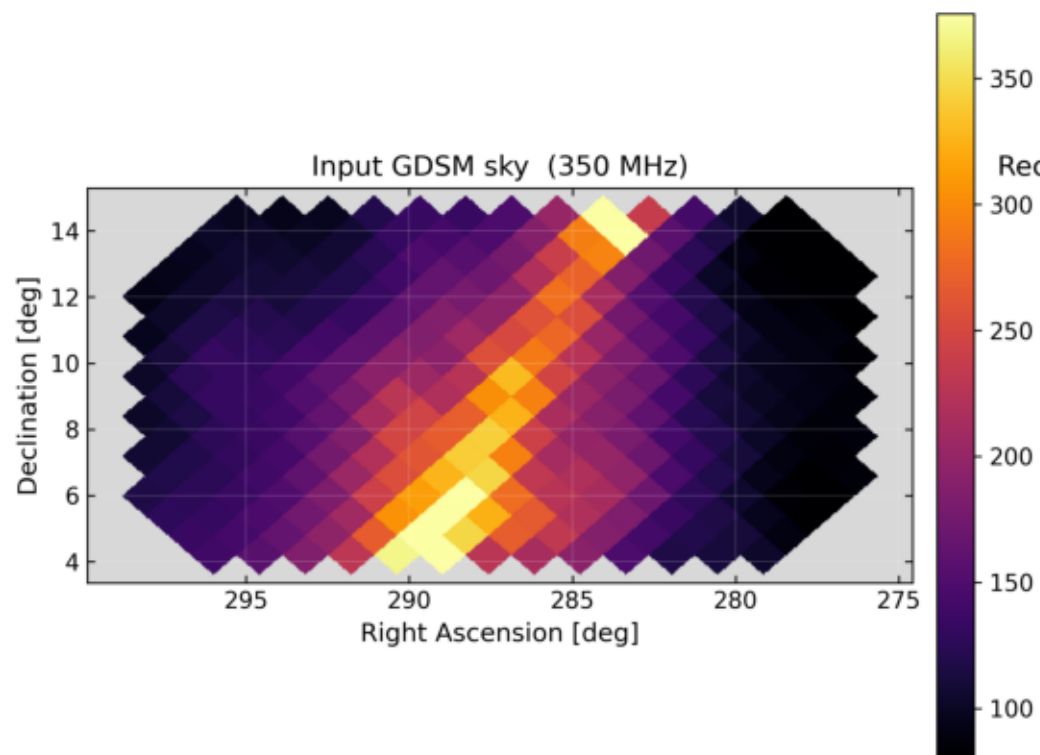
map_ska_galplane_gauss_HP.png



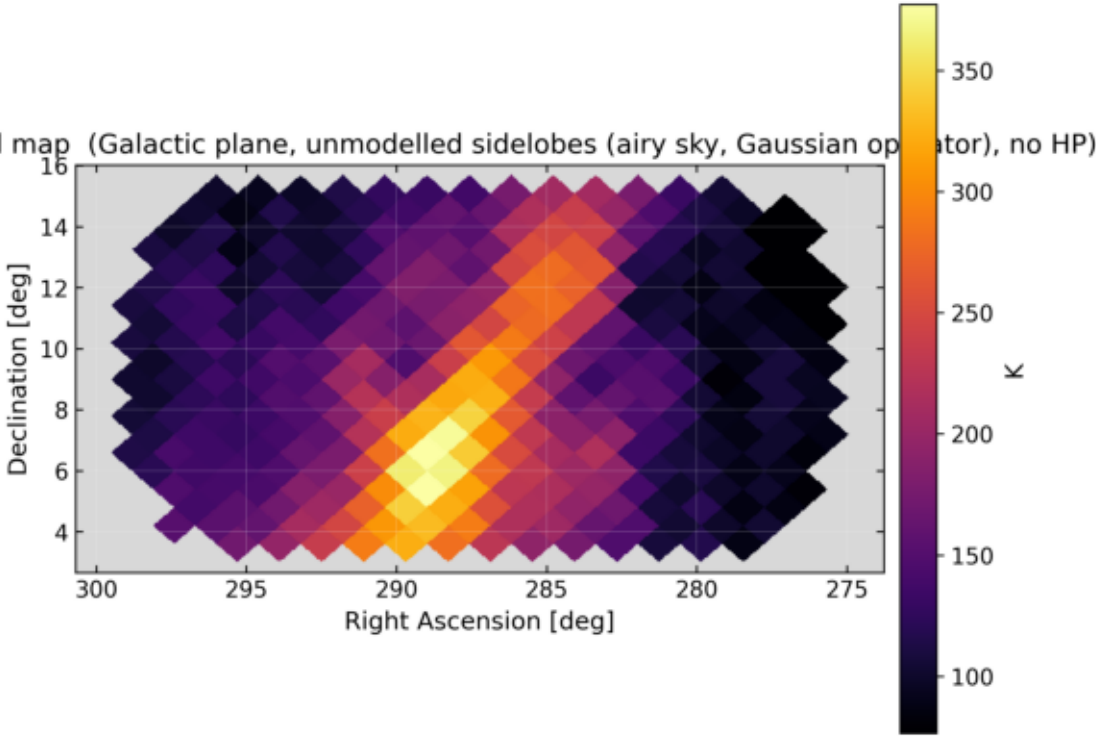
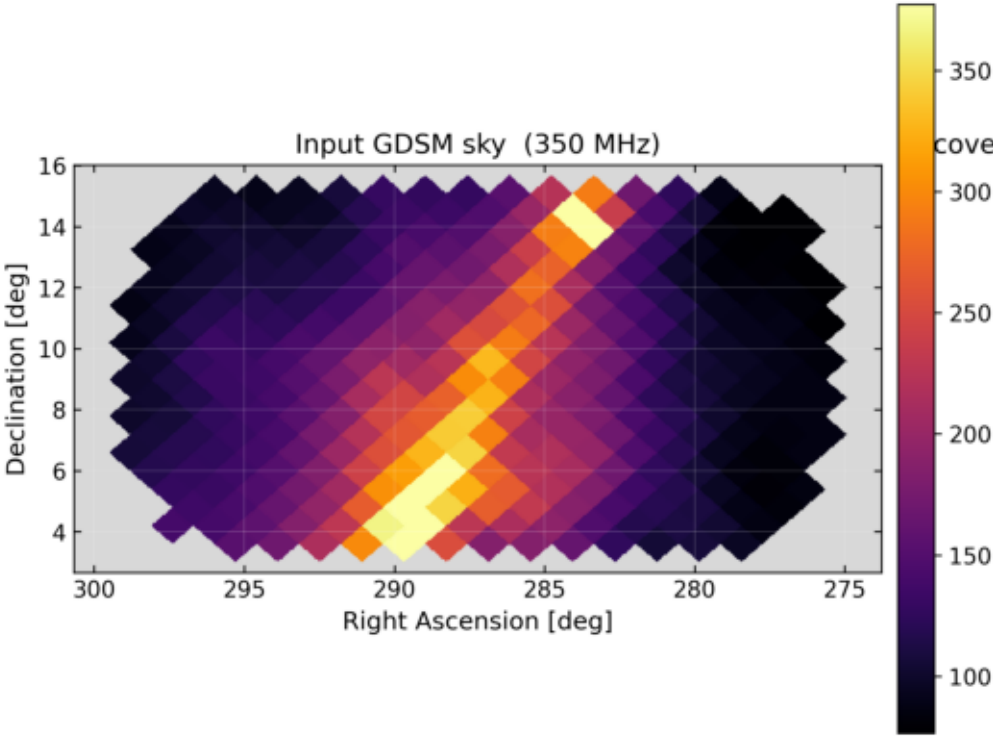
map_ska_galplane_airy_noHP.png



map_ska_galplane_airy_HP.png



map_ska_galplane_mismatch_noHP.png



map_ska_galplane_mismatch_HP.png

